

Assignment 2 Solutions

Statistical Foundation of Data Science (CSU1658)

Md Ayan Alam (GF202342645)

Problem 1: Zero-Inflated Normal

(a) PMF/PDF and CDF:

PDF (mixture distribution):

$$f_X(x) = \begin{cases} \alpha & \text{at } x = 0 \text{ (point mass)} \\ (1 - \alpha) \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2} & x \neq 0 \end{cases}$$

CDF:

$$F_X(x) = \begin{cases} (1 - \alpha)\Phi\left(\frac{x-\mu}{\sigma}\right) & x < 0 \\ \alpha + (1 - \alpha)\Phi\left(\frac{x-\mu}{\sigma}\right) & x \geq 0 \end{cases}$$

where Φ is standard normal CDF.

(b) Mean and Variance:

Expected value:

$$\begin{aligned} \mathbb{E}[X] &= 0 \cdot \alpha + \mu \cdot (1 - \alpha) \\ &= \boxed{(1 - \alpha)\mu} \end{aligned}$$

For variance, compute $\mathbb{E}[X^2]$:

$$\mathbb{E}[X^2] = (1 - \alpha)(\sigma^2 + \mu^2)$$

Then:

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= (1 - \alpha)(\sigma^2 + \mu^2) - (1 - \alpha)^2\mu^2 \\ &= (1 - \alpha)\sigma^2 + (1 - \alpha)\alpha\mu^2 \\ &= \boxed{(1 - \alpha)(\sigma^2 + \alpha\mu^2)} \end{aligned}$$

(c) Numerical: $\alpha = 0.3, \mu = 2, \sigma = 1$

$$\begin{aligned} \mathbb{P}(X \leq 0) &= \mathbb{P}(X = 0) + \mathbb{P}(X < 0) \\ &= \alpha + (1 - \alpha)\Phi\left(\frac{-\mu}{\sigma}\right) \\ &= 0.3 + 0.7 \cdot \Phi(-2) \\ &= 0.3 + 0.7(0.0228) \\ &= \boxed{0.316} \end{aligned}$$

Problem 2: Binomial & Normal Approximation

Given: $X \sim \text{Bin}(10, 0.3)$

(a) Exact:

$$\mathbb{P}(X \geq 6) = \sum_{k=6}^{10} \binom{10}{k} (0.3)^k (0.7)^{10-k}$$

Calculations:

$$P(X = 6) = 210(0.3)^6(0.7)^4 = 0.0368$$

$$P(X = 7) = 120(0.3)^7(0.7)^3 = 0.0090$$

$$P(X = 8) = 45(0.3)^8(0.7)^2 = 0.0014$$

$$P(X = 9) = 10(0.3)^9(0.7)^1 = 0.0001$$

$$P(X = 10) = 1(0.3)^{10} = 0.000006$$

$$\text{Sum: } \boxed{\mathbb{P}(X \geq 6) = 0.0473}$$

(b) Normal Approximation:

Parameters: $\mu = np = 3, \sigma^2 = np(1 - p) = 2.1, \sigma = 1.449$

With continuity correction:

$$\begin{aligned} \mathbb{P}(X \geq 6) &\approx \mathbb{P}(Y > 5.5) \\ &= \mathbb{P}\left(Z > \frac{5.5 - 3}{1.449}\right) \\ &= \mathbb{P}(Z > 1.726) \\ &= 1 - 0.9582 = \boxed{0.0418} \end{aligned}$$

Error: $|0.0473 - 0.0418| = 0.0055$ (11.6%)

Problem 3: Poisson Thinning

(a) Thinning Property:

Given $N \sim \text{Poisson}(\lambda)$, each event tagged "A" with prob. q .

$$\begin{aligned} \mathbb{P}(K = k) &= \sum_{n=k}^{\infty} \binom{n}{k} q^k (1 - q)^{n-k} \frac{\lambda^n e^{-\lambda}}{n!} \\ &= \frac{q^k e^{-\lambda}}{k!} \sum_{n=k}^{\infty} \frac{\lambda^n (1 - q)^{n-k}}{(n - k)!} \end{aligned}$$

Let $m = n - k$:

$$\begin{aligned} &= \frac{(\lambda q)^k e^{-\lambda}}{k!} \sum_{m=0}^{\infty} \frac{[\lambda(1 - q)]^m}{m!} \\ &= \frac{(\lambda q)^k e^{-\lambda}}{k!} \cdot e^{\lambda(1 - q)} \\ &= \frac{(\lambda q)^k}{k!} e^{-\lambda q} \end{aligned}$$

$$\text{Thus: } \boxed{K \sim \text{Poisson}(\lambda q)}$$

(b) Conditional Distribution:

Given $N = n$, each of n events independently tagged "A" with prob. q :

$$K \mid N = n \sim \text{Bin}(n, q)$$

(c) **Numerical:** $\lambda = 12, q = 0.2$, so $\lambda q = 2.4$

$$\mathbb{P}(K = 0) = e^{-2.4} = \boxed{0.0907}$$

$$\begin{aligned}\mathbb{P}(K \leq 2) &= e^{-2.4} \left(1 + 2.4 + \frac{2.4^2}{2} \right) \\ &= 0.0907(1 + 2.4 + 2.88) \\ &= \boxed{0.570}\end{aligned}$$

Problem 4: Data Standardization

Data: $x = (3, 7, 7, 19, 21)$

Basic statistics:

$$\mu = \frac{57}{5} = 11.4, \min = 3, \max = 21, \text{med} = 7$$

$$\begin{aligned}\sigma^2 &= \frac{1}{5} [70.56 + 19.36 + 19.36 + 57.76 + 92.16] \\ &= 51.84, \quad \sigma = 7.2\end{aligned}$$

$$Q_1 = 7, Q_3 = 19, \text{IQR} = 12$$

(a) **Z-scores:** $Z_i = \frac{x_i - 11.4}{7.2}$

$$\begin{aligned}Z_1 &= \frac{-8.4}{7.2} = -1.167, \quad Z_2 = Z_3 = \frac{-4.4}{7.2} = -0.611 \\ Z_4 &= \frac{7.6}{7.2} = 1.056, \quad Z_5 = \frac{9.6}{7.2} = 1.333\end{aligned}$$

$$\text{Verify } \bar{Z} = 0: \frac{1}{5}(-1.167 - 0.611 - 0.611 + 1.056 + 1.333) = 0$$

$$\text{Verify } \text{Var}(Z) = 1: \frac{1}{5}(1.362 + 0.373 + 0.373 + 1.115 + 1.777) = 1$$

(b) **Correlation invariance:**

$$\begin{aligned}\rho(X, aY + b) &= \frac{\text{Cov}(X, aY + b)}{\sigma_X \sigma_{aY + b}} \\ &= \frac{a \cdot \text{Cov}(X, Y)}{a \cdot \sigma_X \sigma_Y} = \rho(X, Y)\end{aligned}$$

(c) **Inverse:** $X_i = 11.4 + 7.2Z_i = 3 + 18M_i$

(d) **Min-max:** $M_i = \frac{x_i - 3}{18}$

$$M = (0, 0.222, 0.222, 0.889, 1.0)$$

$$\text{Robust: } R_i = \frac{x_i - 7}{12} = (-0.333, 0, 0, 1.0, 1.167)$$

Problem 5: Binning with Outlier

Data: 199 points $\sim N(0, 1)$, outlier $x_{200} = 8$

$$\sum_{i=1}^{199} x_i = 0, \sum_{i=1}^{199} x_i^2 = 199$$

(a) **Statistics with outlier:**

$$\bar{x} = \frac{0 + 8}{200} = 0.04$$

$$\sum x_i^2 = 199 + 64 = 263$$

$$\begin{aligned}s^2 &= \frac{263}{200} - (0.04)^2 = 1.315 - 0.0016 = 1.3134 \\ s &= 1.146\end{aligned}$$

(b) **Bin widths:**

$$\text{Sturges: } K = \lceil \log_2(200) \rceil + 1 = \lceil 7.64 \rceil + 1 = 9$$

Scott:

$$\begin{aligned}h_S &= 3.5 \cdot s \cdot n^{-1/3} \\ &= 3.5(1.146)(200)^{-1/3} \\ &= 3.5(1.146)(0.1716) = 0.688\end{aligned}$$

FD (using $\text{IQR} \approx 1.349$ for standard normal):

$$\begin{aligned}h_{FD} &= 2 \cdot \text{IQR} \cdot n^{-1/3} \\ &= 2(1.349)(0.1716) = 0.463\end{aligned}$$

(c) **Bins on $[-4, 8]$ (range=12):**

$$\text{Sturges: } 12/1.33 = 9 \text{ bins}$$

$$\text{Scott: } 12/0.688 \approx 17 \text{ bins}$$

$$\text{FD: } 12/0.463 \approx 26 \text{ bins}$$

FD most robust (IQR unaffected by single outlier).

Problem 6: Logistic Regression (2x2)

Data: $a = 18, b = 12, c = 30, d = 10$

(a) **Coefficients:**

$$\begin{aligned}\hat{\beta}_0 &= \log \frac{a}{b} = \log \frac{18}{12} = \log \frac{3}{2} = \ln 3 - \ln 2 \\ &= 1.0986 - 0.6931 = \boxed{0.4055} \\ \hat{\beta}_1 &= \log \frac{bc}{ad} = \log \frac{12 \times 30}{18 \times 10} = \log \frac{360}{180} \\ &= \log 2 = \boxed{0.6931}\end{aligned}$$

$$\text{Odds ratio: } \exp(0.6931) = \boxed{2.0}$$

(b) **Variance and CI:**

$$\begin{aligned}\text{Var}(\hat{\beta}_1) &= \frac{1}{18} + \frac{1}{12} + \frac{1}{30} + \frac{1}{10} \\ &= 0.0556 + 0.0833 + 0.0333 + 0.1000 = \frac{49}{180} \\ &= 0.2722 \\ SE(\hat{\beta}_1) &= \sqrt{0.2722} = 0.5218\end{aligned}$$

Margin of error: $ME = 1.96 \times 0.5218 = 1.0227$

CI for β_1 : $[0.6931 - 1.0227, 0.6931 + 1.0227] = [-0.330, 1.716]$

95% CI for OR:

$$[\exp(-0.330), \exp(1.716)] = [0.719, 5.563] = \boxed{[0.72, 5.56]}$$

Problem 7: Within-Subject ANOVA

Data (5 subjects $\times$ 3 conditions):

Subj	C1	C2	C3	Mean
1	10	12	13	35/3=11.67
2	9	11	12	32/3=10.67
3	8	10	11	29/3=9.67
4	9	10	12	31/3=10.33
5	11	12	14	37/3=12.33
Mean	47/5=9.4	55/5=11	62/5=12.4	164/15=10.933

Sum of Squares:

$$SS_{Cond} = 5[(9.4 - 10.933)^2 + (11 - 10.933)^2 + (12.4 - 10.933)^2] \\ = 5[(-1.533)^2 + (0.067)^2 + (1.467)^2] = 5[2.351 + 0.004 + 2.152] = 5 \times 4.507 = 22.533$$

$$SS_{Subj} = 3[(11.667 - 10.933)^2 + (10.667 - 10.933)^2 + (9.667 - 10.933)^2 \\ + (10.333 - 10.933)^2 + (12.333 - 10.933)^2] = 3[0.539 + 0.071 + 1.603 + 0.360 + 1.960] = 3 \times 4.533 = 13.600$$

$$SS_{Total} = \sum (x_{ij} - 10.933)^2: 15 \text{ terms} \Rightarrow SS_{Total} = 36.930$$

$$SS_{Error} = 36.930 - 22.533 - 13.600 = 0.797$$

Mean Squares and F:

$$MS_{Cond} = 22.533/2 = 11.267, MS_{Error} = 0.797/8 = 0.100$$

$$F = 11.267/0.100 = \boxed{112.67}$$

$$F_{0.05, 2, 8} = 4.46 \rightarrow \text{Reject } H_0 \text{ (conditions differ, } p < 0.001)$$

Problem 8: Impurity Measures

(a) Prove $H(p) \geq G(p)$:

$$H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$$

$$G(p) = 2p(1-p)$$

$$\text{At } p = 0.5: H(0.5) = 1, G(0.5) = 0.5$$

$$\text{By calculus: } H(p) - G(p) \geq 0 \text{ for all } p \in [0, 1]$$

(b) Prove $G(p) \geq E(p)$:

$$E(p) = \min\{p, 1-p\}$$

$$\text{For } p \leq 0.5: G(p) = 2p(1-p) \geq p = E(p)$$

$$\text{since } 2(1-p) \geq 1 \text{ when } p \leq 0.5.$$

(c) Comparison table:

p	$H(p)$	$G(p)$	$E(p)$
0.1	0.469	0.18	0.1
0.3	0.881	0.42	0.3
0.5	1.000	0.50	0.5

Problem 9: Information Gain

12 samples: 6 positive, 6 negative

Parent impurity:

$$H(S) = -0.5 \log_2(0.5) - 0.5 \log_2(0.5) = 1.0$$

$$G(S) = 2(0.5)(0.5) = 0.5$$

Attribute A (v_1 : 4/1, v_2 : 1/2, v_3 : 1/3):

Weighted entropy:

$$H(S|A) = \frac{5}{12}H(0.8) + \frac{3}{12}H(0.33) + \frac{4}{12}H(0.25) \\ = \frac{5}{12}(0.722) + \frac{3}{12}(0.918) + \frac{4}{12}(0.811) \\ = 0.801$$

$$IG(S; A) = 1.0 - 0.801 = \boxed{0.199}$$

Weighted Gini:

$$G(S|A) = \frac{5}{12}(0.32) + \frac{3}{12}(0.444) + \frac{4}{12}(0.375) \\ = 0.369$$

$$\text{Gini decrease: } 0.5 - 0.369 = \boxed{0.131}$$

Attribute B (b_0 : 4/2, b_1 : 2/4):

$$H(S|B) = 0.5H(0.667) + 0.5H(0.333) = 0.918$$

$$IG(S; B) = 1.0 - 0.918 = \boxed{0.082}$$

$$G(S|B) = 0.5(0.444) + 0.5(0.444) = 0.444$$

$$\text{Gini decrease: } 0.5 - 0.444 = \boxed{0.056}$$

Conclusion: Both ID3 and CART choose **Attribute A** (higher gain).

Problem 10: Simple Linear Regression

Data: (0, 1.0), (1, 2.9), (2, 5.1), (3, 5.9), (4, 9.2)

(a) OLS Estimates:

$$\bar{x} = 2, \bar{y} = 4.82$$

$$S_{xx} = 10, S_{xy} = 19.4, S_{yy} = 38.708$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{19.4}{10} = \boxed{1.94}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 4.82 - 3.88 = \boxed{0.94}$$

$$\text{Fitted line: } \hat{y} = 0.94 + 1.94x$$

(b) Fit Statistics:

x	y	$\hat{y}$	e	e^2
0	1.0	0.94	0.06	0.0036
1	2.9	2.88	0.02	0.0004
2	5.1	4.82	0.28	0.0784
3	5.9	6.76	-0.86	0.7396
4	9.2	8.70	0.50	0.2500

$$SSE = 1.072, SSR = 37.636$$

$$R^2 = \frac{SSR}{S_{yy}} = \frac{37.636}{38.708} = \boxed{0.9723}$$

$$s = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{1.072}{3}} = \boxed{0.598}$$

(c) Hypothesis Test:

$$H_0 : \beta_1 = 0 \text{ vs } H_a : \beta_1 \neq 0$$

$$SE(\hat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}} = \frac{0.598}{3.162} = 0.189$$

$$t = \frac{1.94}{0.189} = \boxed{10.265}$$

$$df = 3, t_{0.025,3} = 3.182$$

Reject H_0 ($p \approx 0.002$)

$$95\% \text{ CI: } 1.94 \pm 3.182(0.189) = \boxed{(1.339, 2.541)}$$

(d) Prediction Interval at $x_0 = 5$:

$$\hat{y}_0 = 0.94 + 1.94(5) = \boxed{10.64}$$

$$SE_{pred} = s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

$$= 0.598 \sqrt{1 + 0.2 + 0.9} = \boxed{0.866}$$

$$95\% \text{ PI: } 10.64 \pm 3.182(0.866) = \boxed{(7.88, 13.40)}$$

Note: Extrapolation beyond $[0, 4]$.